

Design Review: Canine Crash Test Dummy

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Design Review: Canine Crash Test Dummy

This project is a canine crash test dummy that is to be used on sled testing canine harnesses. There are typically standards that are set for all types of existing crash tests however canine harness crash testing has none. In June of 2019 a certain American Society for Testing and Materials (ASTM) committee was working on developing a standard for canine harness crash tests. As of today, there is no official standard for testing on these types of restraint systems, meaning strictly harnesses that aim to reduce the number of accidents where a pet is ejected from a vehicle, reduce the number of accidents where the pet becomes a projectile and hits another passenger in the vehicle, offer a safety restraint for unrestrained pets in the vehicle, and to help control the pet after a collision. The ASTM committee proposed that the standard they were trying to develop would be similar to the standards for assessment of child restraints, FMVSS 213. These standards would have a sled testing configuration with a standard bench seat, and testing would be performed under specifications compatible to those mentioned in the FMVSS 213 standards. A problem with the proposal was that there was not enough justification for some of the aspects of the crash tests such as the size of the harness, the center of mass, and the impact of the skin and fur of the canine on the testing.

This capstone project is directed to further define some of these parameters so that a second proposal can be submitted, and a uniform test can be established for canines that fall into certain size and weight categories in harness type restraint systems in moving automobiles. The long-term objective is to reduce the abuse of pets that are subjected to automobile collisions. Because the size and weight of dogs have such a large range, there are several different sizes of harnesses that are available, each of which needs its own standard developed during crash testing. This means that the parameters of the project dummy must fall into the correct size and

weight ranges of a medium sized dog. These parameters include the center of mass, overall weight, overall height, overall length, and chest circumference. The model that was created was designed after the Beagle breed of canine, which is classified as a medium sized dog according to the standards of the American Kennel Club. All of the parameters mentioned previously must fall into the specified ranges of a medium sized dog to ensure that the dummy will behave appropriately when it is tested on a medium sized harness designed for a medium sized dog.

Design Requirements - Christian & Sebastian

The team reached out to two different customers, including Exponent, the company the team is doing the capstone project to. From the information gathered from meeting the customers, the team was able to determine various customer requirements and identify different engineering standards and design specifications for the customer requirements. The following table provides different functional and customer requirements with their corresponding engineering standards and design specifications.

Functional Requirements

Provide details on the functional requirements and refer to Table 1 in the discussion.

Table 1: Functional Requirements and Design Specifications

Functional Requirements	Customer Requirement	Engineering Standard	Design Specifications
Design must have a center of mass located in the middle upper chest cavity while still maintaining an overall weight of 25 lbs - 30 lbs	Ensure model weights at least 25 lbs.	ASTM test specifications.	Model can weigh up to 30 lbs but must be at least 25 lbs.
Design must be durable enough to withstand multiple	Use metal for the bone structure, foam for the muscle	ASTM D - 3574 - 17	Ensure the selected foam is dense enough and flexible to wrap

sled tests at speeds up to 30 mph without deforming	resemblance and canvas for the skin of the model.		around the canine bond structure and provide a feel of the real thing.
No Instrumentation is to be included	Avoid instrumentation on the model.	IEEE standards are not applicable.	Do not use instrumentation on the model.
Height of design must be between 45 cm - 48 cm and a length between 55 cm - 58 cm. Design must have a chest girth between 22.5 in - 28 in	Fit appropriately on a seat belt harness	FMVSS - 209	The model must have appropriate dimensions to fit in a harness; the harness must perform to the 209 regulations.
Design must be durable enough to withstand multiple sled tests at speeds up to 30 mph without deforming	Model must withstand high force	FMVSS - 210	Model in the harness must withstand a 5000 lbs. force.
The adhesive used to attach the foam and canvas to the metal bone structure should not be acidic enough to dissolve the foam or canvas	Adhesive used on the model must work effectively and safely secure the materials together.	ASTM D2651 - 01	The adhesive should be able to secure the foam onto metal throughout multiple crash tests and the foam should not fall out when the canvas fabric is removed.
At no point during crash testing should the metal bone structure be protruding from the canvas layer representing skin/fur	Selected canvas must not tear under repeated friction between the fabric and seatbelt.	ASTM D3884 - 01	Canvas fabric should have a high durability to prevent disturbing or adding errors to the crash tests.
Range of motion should mimic that of an actual medium sized canine	Joints of dog should allow it to be configured in both seated and laying positions without the range of motion being impeded by the foam or canvas layers		Resistance of joints should be adjustable

Sheet metal should be thin enough to be bent to the desired shape and chest circumference of a medium sized dog	Chest of design should be able to properly fit a harness meant for a medium sized dog		Sheet metal used should not be thicker than 16-gauge sheet metal
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Design Specifications

Exponent's representative, Bruce Miller, provided a lot of feedback on the current design of the canine crash test dummy the team had designed. Bruce Miller provided the following suggestions for the design of the canine crash test dummy: Make sure to keep the canine crash test dummy at least at 25 lbs. in order to meet the ASTM specifications, avoid friction between the canvas of the model and the harness, design front paws to prevent damaging the car's seat and perforations on the foam and canvas of the model and select a foam that would resemble the feel of a dog . From the provided suggestions, the team made several changes to the design of the canine crash test dummy and the material selection for its construction. In order to keep the model at, at least 25 lbs. the team decided to add weights to the model to the chest cavity in case it was underweight; this The team also made changes to the front paws of the model to keep the car's seat protected and avoid any perforations on the foam and canvas. Lastly, the team made several changes to the material selection for the construction of the canine crash test dummy. The foam selected polyurethane foam sheet since it provides firmness and pliability to the skin to resemble the real thing.

Making sure the model weights at least 25 lbs. will help the team meet the ASTM test specifications. The polyurethane foam sheet is a foam that is dense and flexible enough to wrap around the canine bone structure allowing the team to meet the ASTM D-3574-17 standards. The design of the canine crash test dummy follows the exact measurements of the beagle, therefore it

will be able to fit properly on the harness; it will meet the FMVSS-209 standard. The team will be getting a harness that fits the model design in order to verify the FMVSS-210 standard. The 8 oz. Gorilla glue that will be used to attach the foam to the bone structure of the canine crash test dummy will allow the team to meet the ASTM D2651-01 standard since this glue is considered one of the strongest adhesives; the foam will not detach once attached to the bone structure using the selected adhesive. Lastly, the 60" x 36" 10 oz. Canvas Cloth will help meet the ASTM D3884-01 standard since this type of fabric is durable. However, the material will be tested to see if it actually is durable . All of these standards, however, will be tested to determine whether the designs and materials selected meet the various engineering standards.

Final Design - Jake

Design 1

The first design was a simple sketch of a Labrador. The goal of this was to brainstorm different joints and points of mass required to resemble the size and shape of a common dog breed.

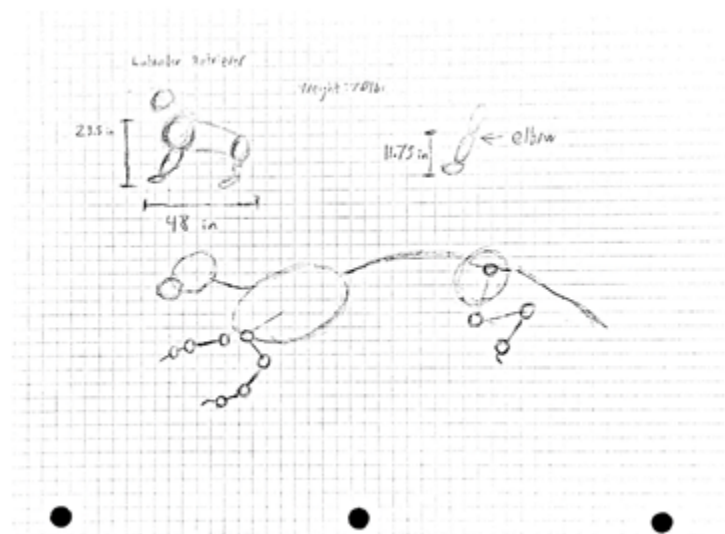


Figure 1. First Design Concept Sketch

Design 2

This design was a quick put together SolidWorks model to figure out the best methods of making each of the parts. No dimensions or weighting was considered in this model.

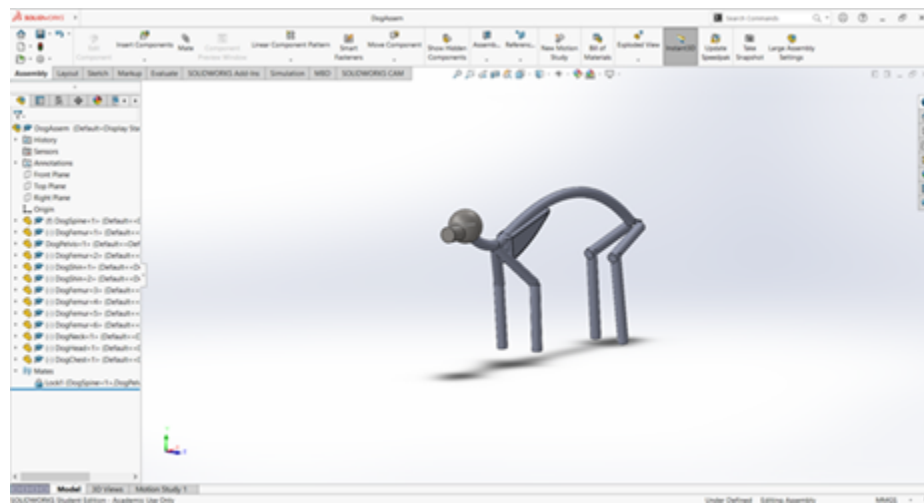


Figure 2. First Design Concept SolidWorks Model

Design 3

This model shows a complete redo of the design from the previous submission. The model now featured working plate joints on the legs and ball and socket joints on the hips and shoulders. The size of the ribcage was also increased noticeably to a more accurate size of a Beagle, which is the final choice for what the dog would be modeled after. The shoulder blades are also a new addition that allowed proper range of motion of the front legs.

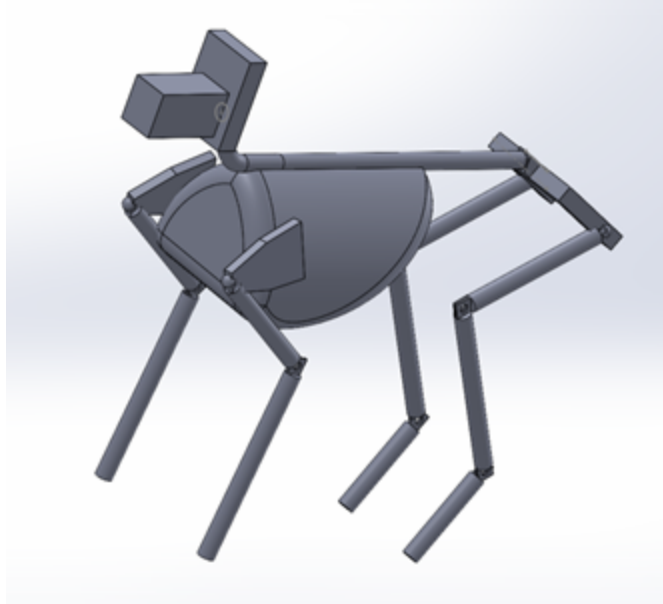


Figure 3. Current Design Concept - Crash Test Dummy Prototype version 2

Design 4

This represents how the design was able to achieve a sitting position with the addition of the joints mentioned previously. The head position was also fixed by straightening out the neck. There is also now a drawing that shows the model is within the range of dimensions for the common beagle.



Figure 4. Current Design Concept - Dog Prototype 2 (Sitting)

Design5

The goal of this design was to solve for the massive weight and center of mass issue. The chest needed to be hollowed out, which caused the model to weigh 28.2 lbs at the time. This fell within the goal range of 20-30 lbs for a beagle. The center of mass was also now accurate to diagrams researched by the team.



Figure 6. Current Design Concept - Dog prototype 2.1

Design 6

The model now includes a modified neck assembly that would be capable of deflection of a real beagle's neck. It is an integrated design that uses real crash test dummy necks. The neck has also been hollowed out for center of mass and weight reasons.



Figure 7. Current Design Concept - Dog prototype 2.2

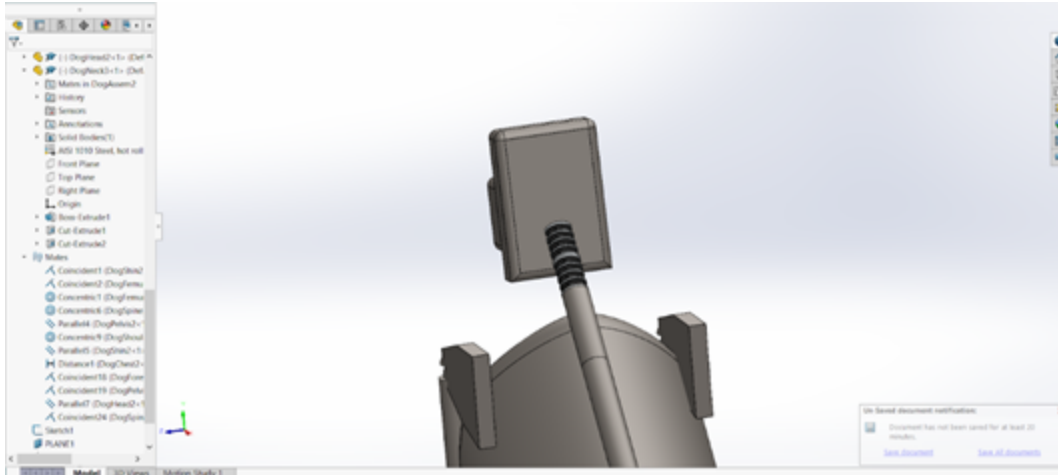


Figure 8. Current Design Concept - Dog prototype 2.2 neck

Design 7

This design goal was to select possible joint models to connect the bone structure together. The joints are crucial to the range of motion. The ball and socket model and knuckle joint were two possible options selected to model.



Figure 9. Current Design Concept - Ball and Socket Joint



Figure 10. Current Design Concept - Plate/Knuckle Joint

Design 8

This design includes a modification to the shoulder position/design, as well as actual models for the ball and socket and plate joints. This orientation will simulate the dog's range of motion as accurate as possible. The socket also includes a possible screw that will be drilled into the outside to allow adjustable friction. The plate joints are a screw, bolt, and two washers. The shoulder is also reinforced with a pipe that will be attached to the spine. The mass of the dog was 28.2 lbs.



Figure 11. Current Design Concept - Dog prototype version 3.0

Design 9

This is the final design submitted last semester. The chest has been swapped out for a 16 gauge sheet metal thickness for a better representation of what is more realistic to accomplish in real-life. This however decreased the weight of the model from 28 to 14.8 lbs. This can be solved through placements of extra weight inside the chest. There is also a drawing present that shows the model is within the desired range of dimensions for the common Beagle.



Figure 12. Current Design Concept - Dog prototype Version 3.1

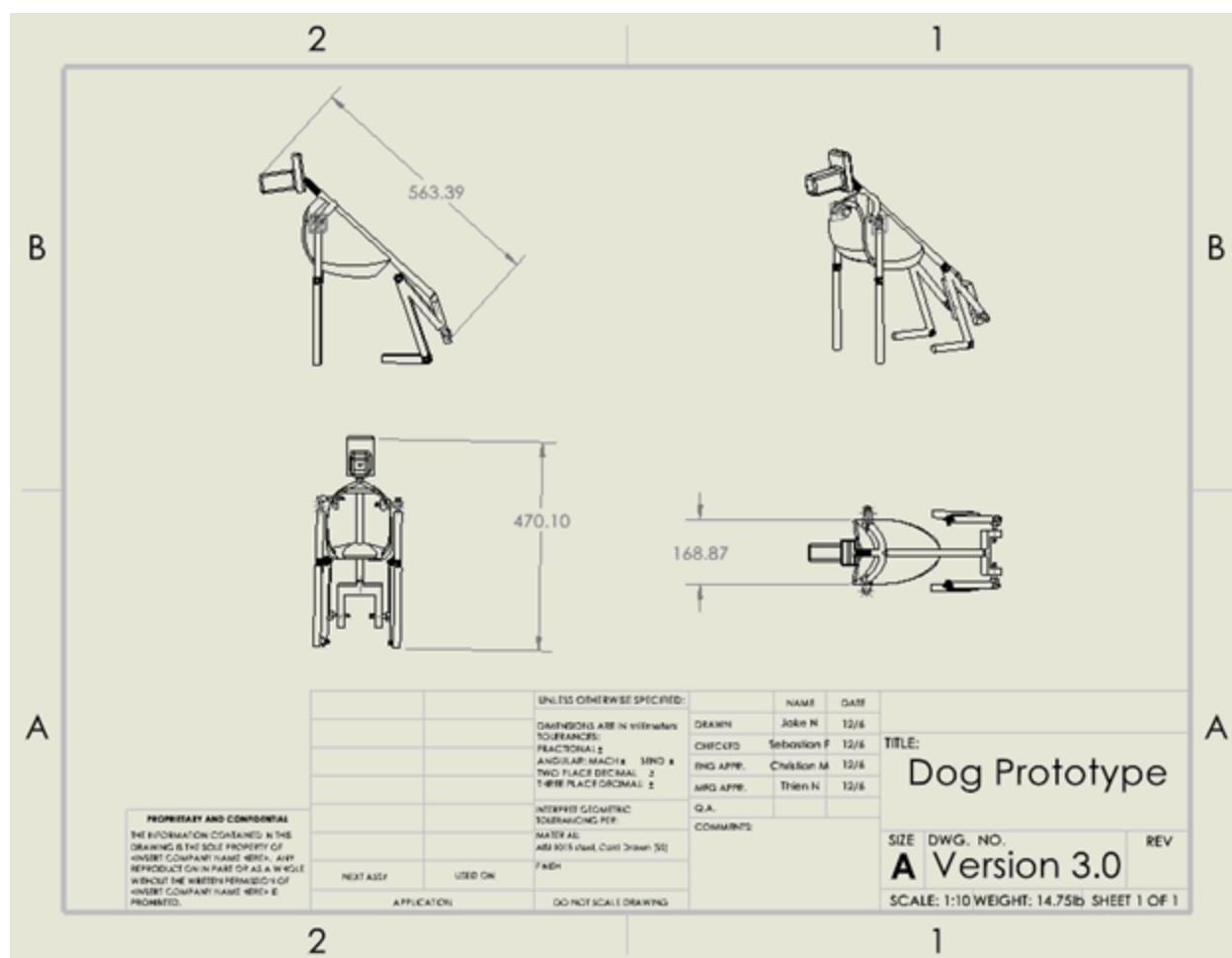


Figure 13. Current Design Concept - Dog prototype Version 3.1 - Drawing

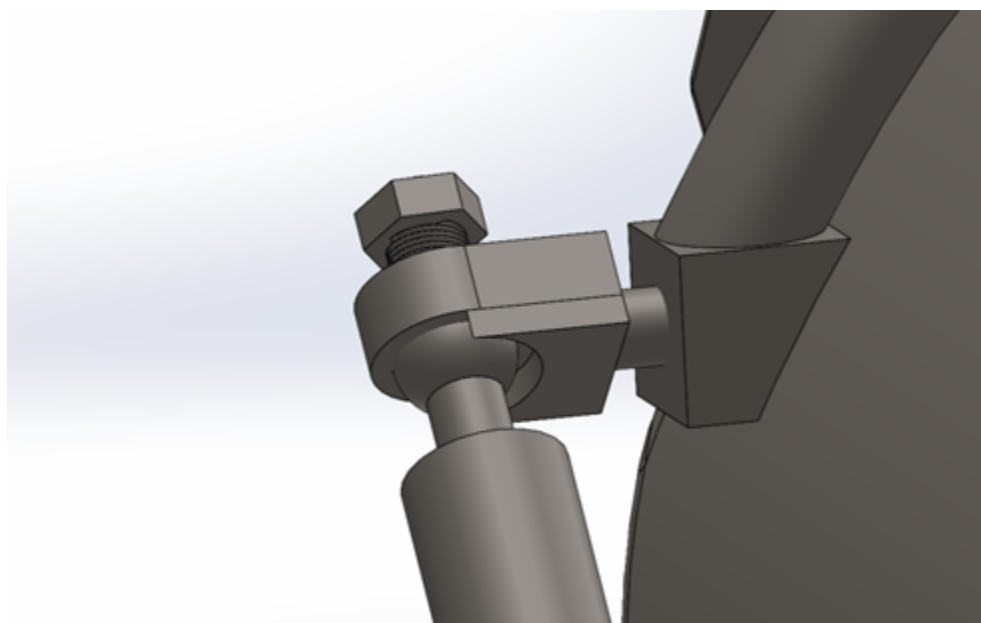


Figure 14. Current Design Concept - Dog prototype Version 3.1 – Ball & Socket Joint

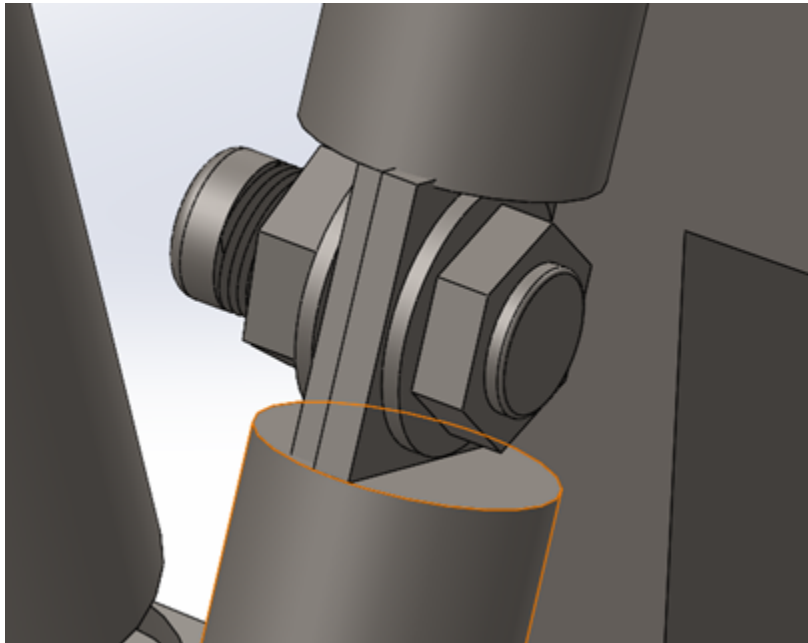


Figure 15. Current Design Concept - Dog prototype Version 3.1 – Plate Joint

Design 11

The final design of the crash test dummy dog is pictured below. One final adjustment was made by adding two front paws to the front legs of the dog. This will prevent the sharp corners from puncturing the seat of the vehicle being tested. These balls will be steel or rubber.



Figure 16. Current Design Concept - Dog prototype Version 3.2

The outer shell of the metal skeleton of the dog will consist of canvas fabric skin and foam to act as muscle filler. The team has been at work in the engineering shop the past week and has completed the necessary cutting for now. The next steps are to move onto welding and joint construction. The team also plans on cutting the sheet metal chest cavity at GCU and possibly outsourcing the bending of the chest to an external company.

Mechanical Design

There are no software or electrical components in this design. The body of the dog features an all-steel assembly with mechanical joints. Below are the various parts necessary that the team has currently begun constructing.

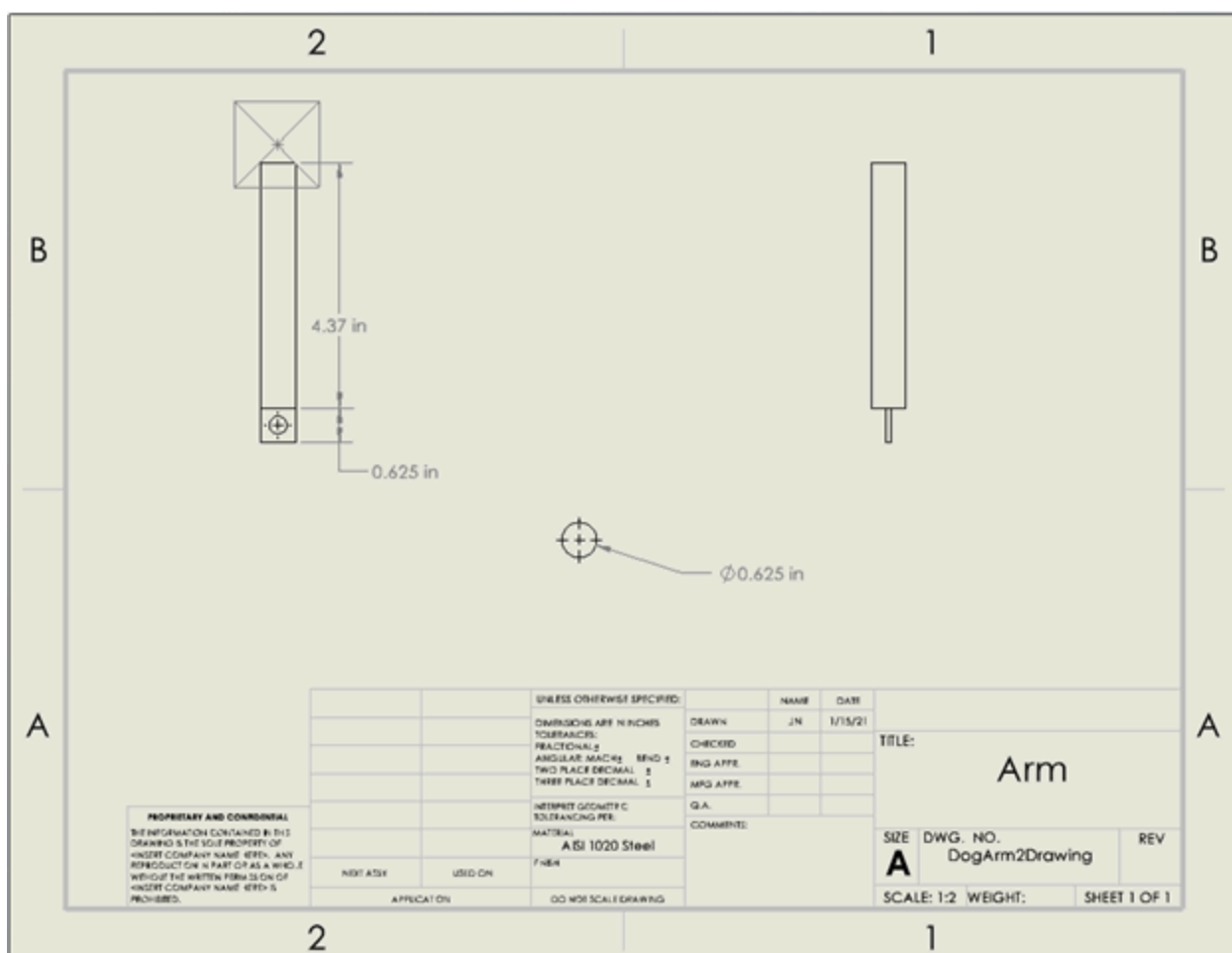


Figure 17. Dog Prototype – Arm Drawing – 2 Total

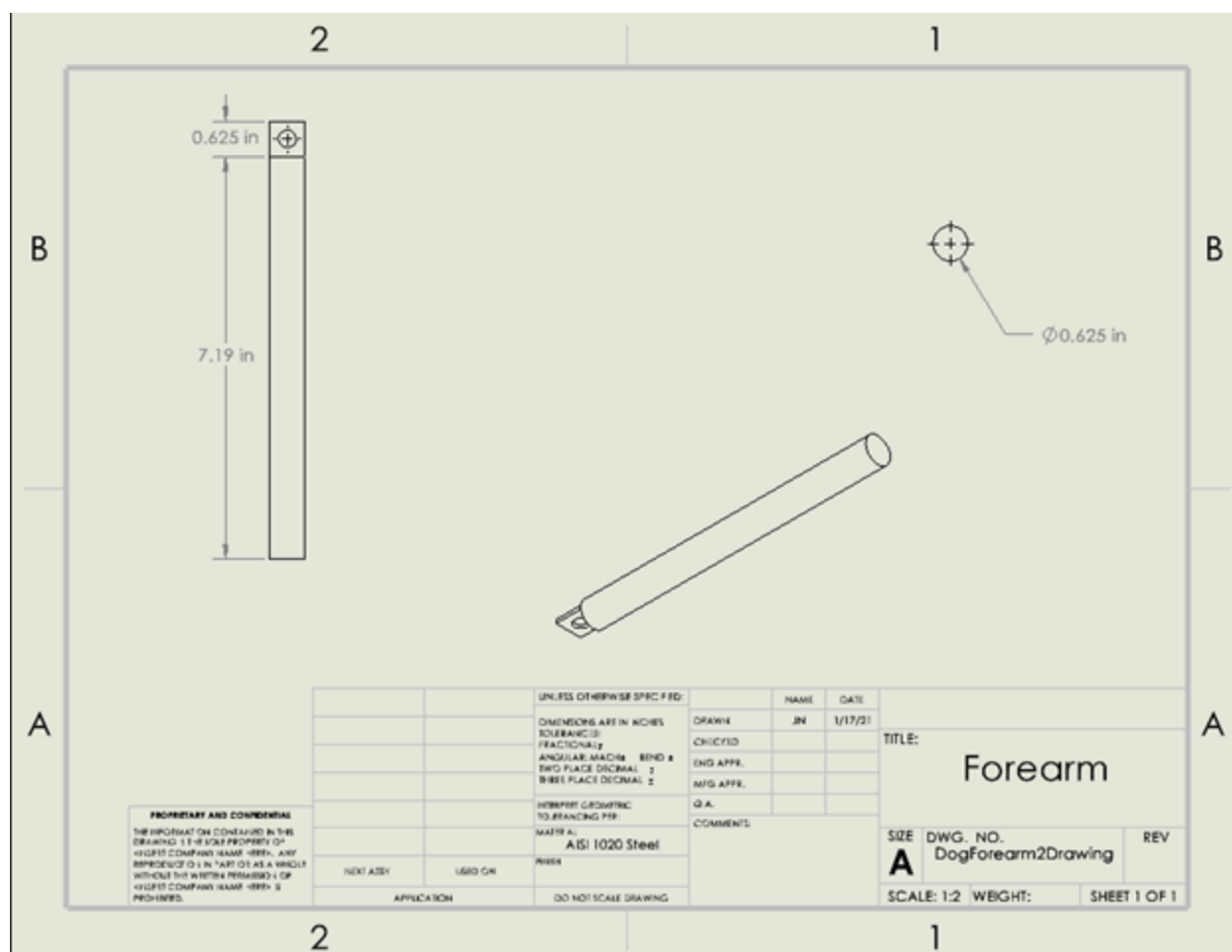


Figure 18. Dog Prototype – Forearm Drawing – 2 Total

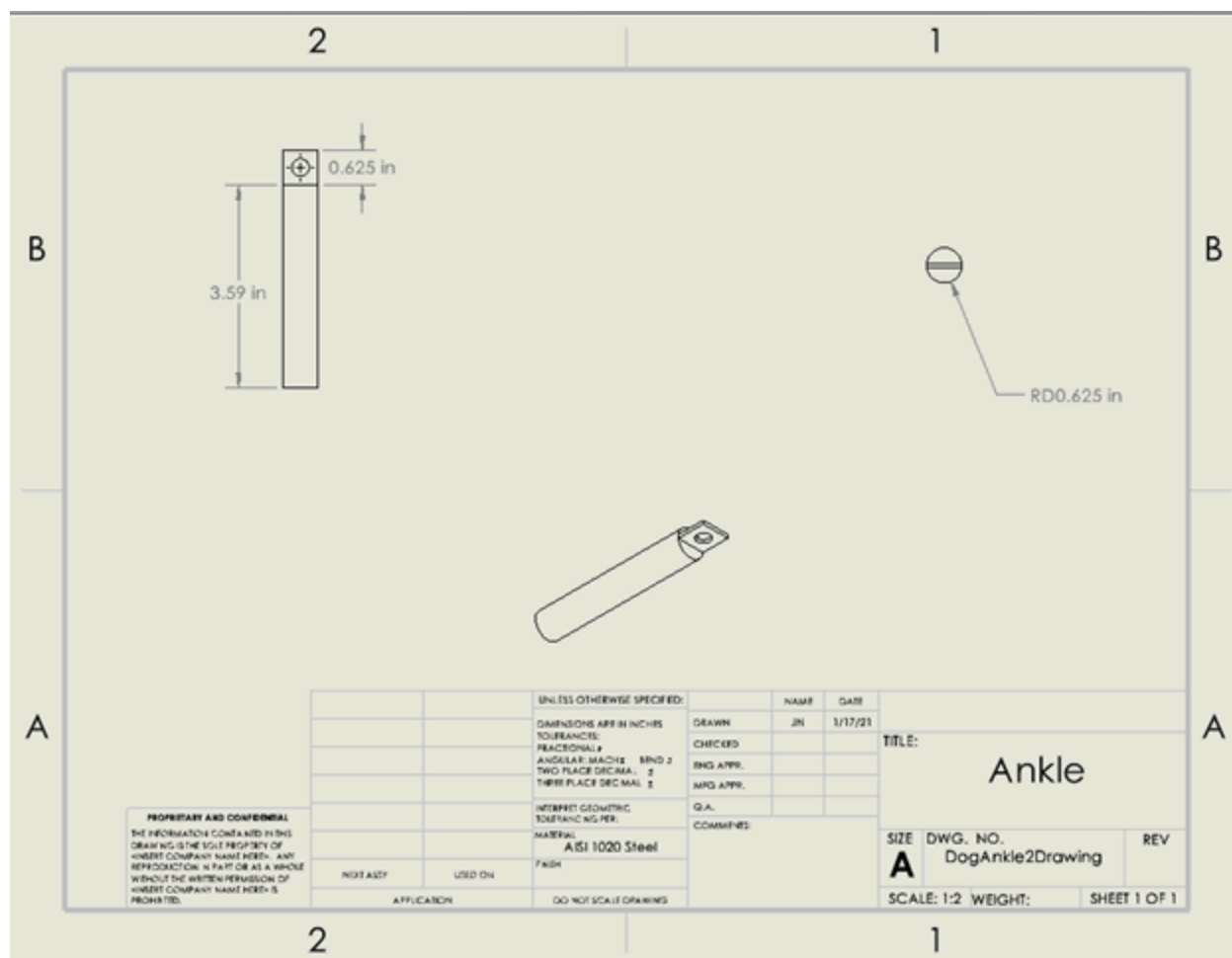


Figure 19. Dog Prototype – Ankle Drawing – 2 Total

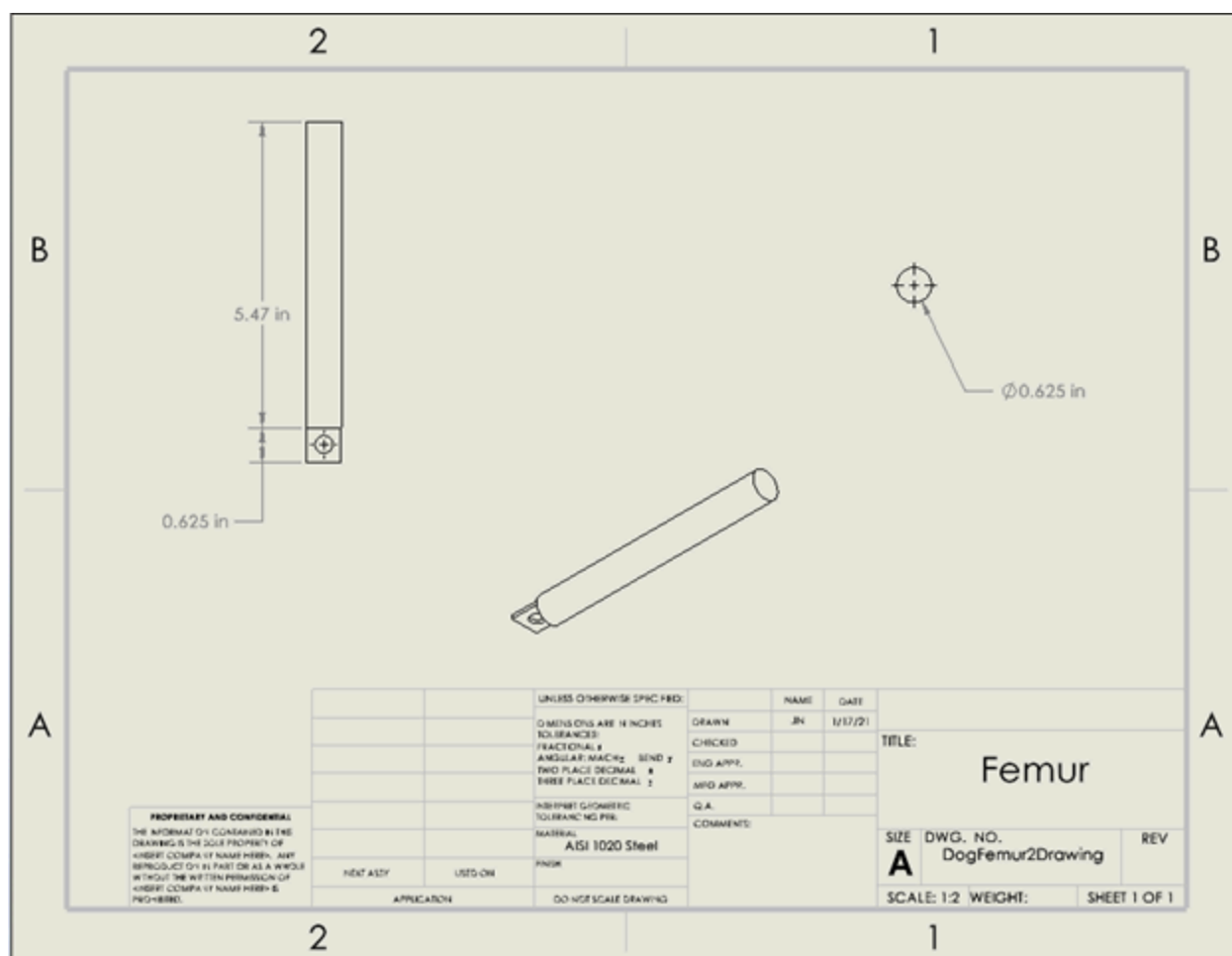


Figure 20. Dog Prototype – Femur Drawing – 2 Total

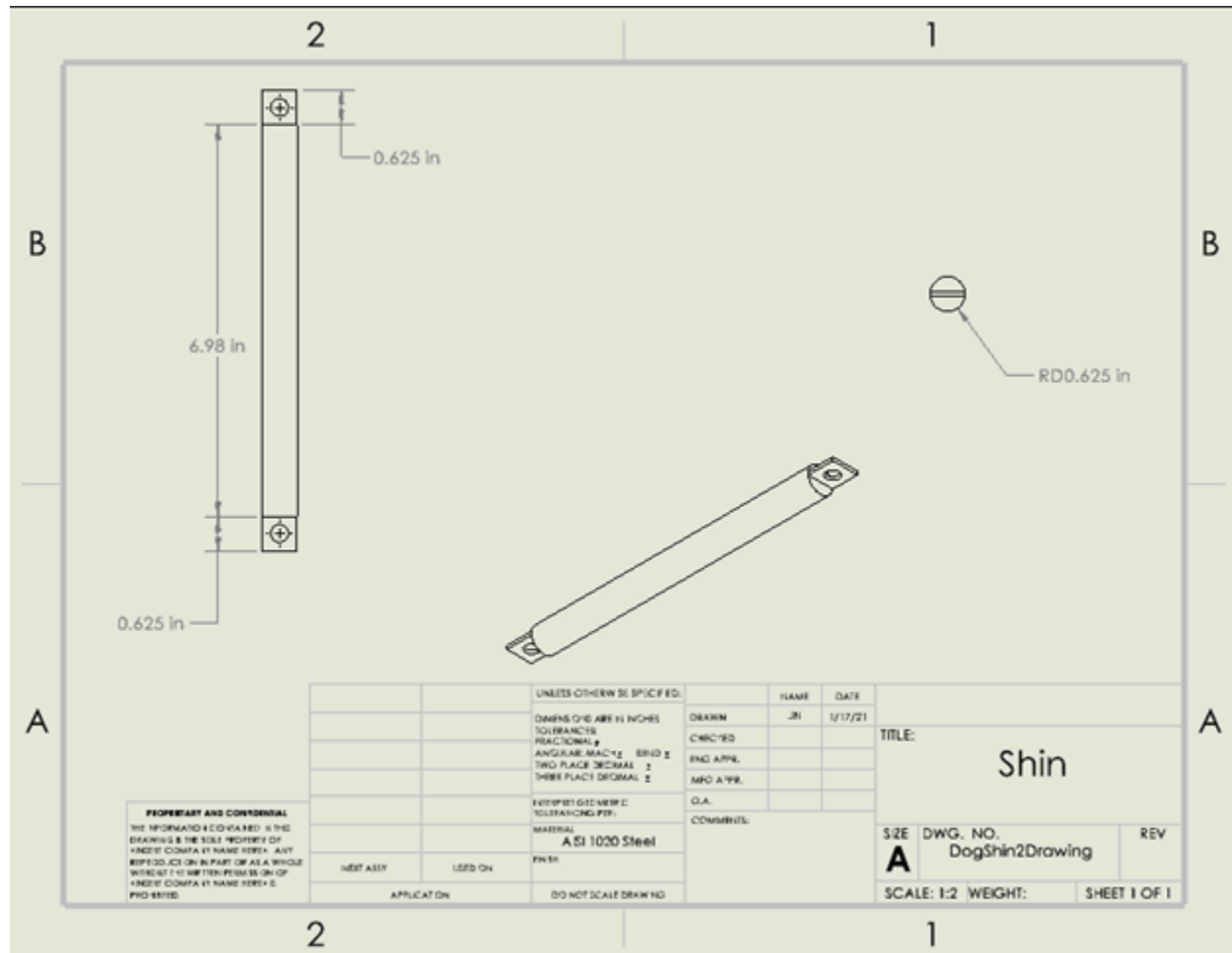


Figure 21. Dog Prototype – Shin Drawing – 2 Total

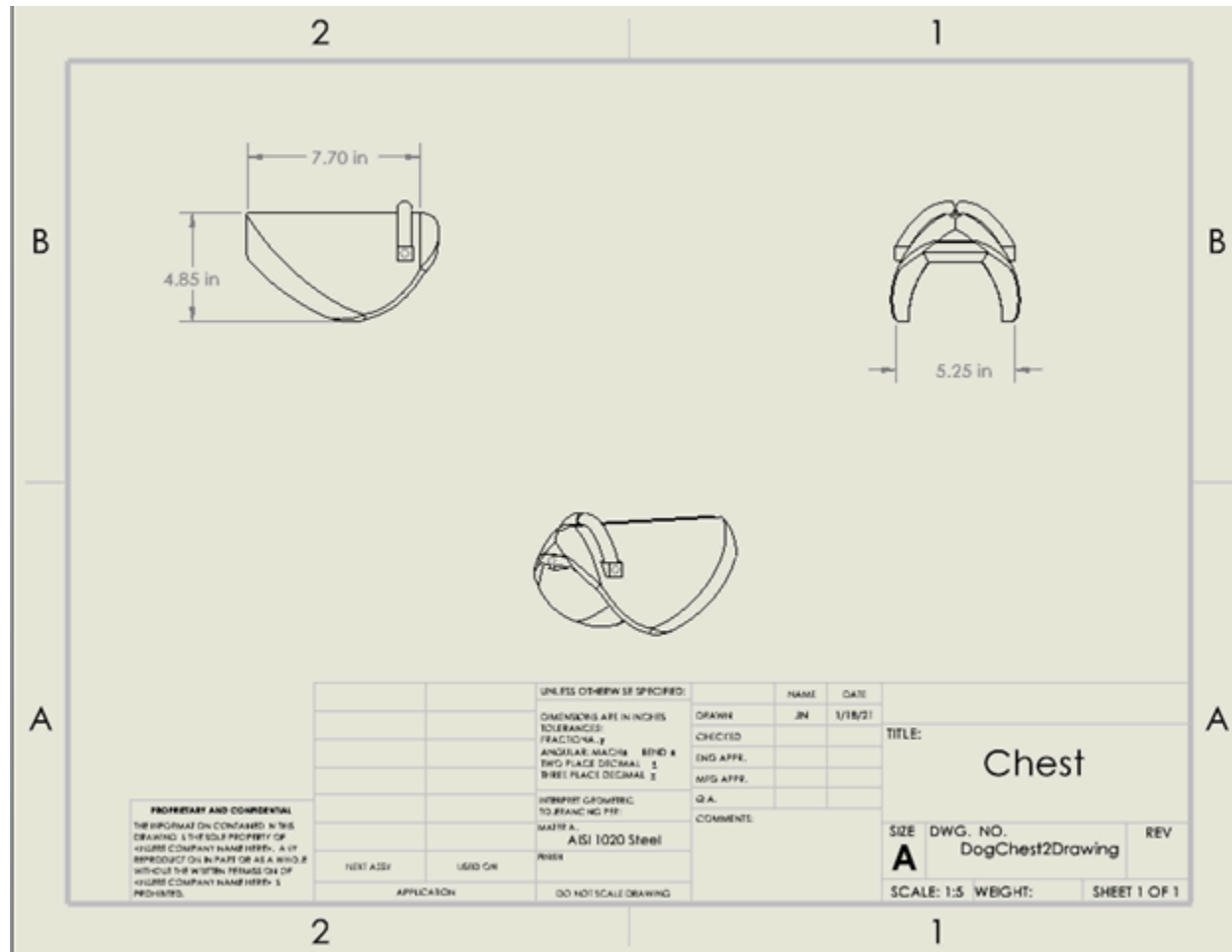


Figure 22. Dog Prototype – Chest Drawing – 1 Total

Figure 23. Dog Prototype – Spine Drawing – 1 Total

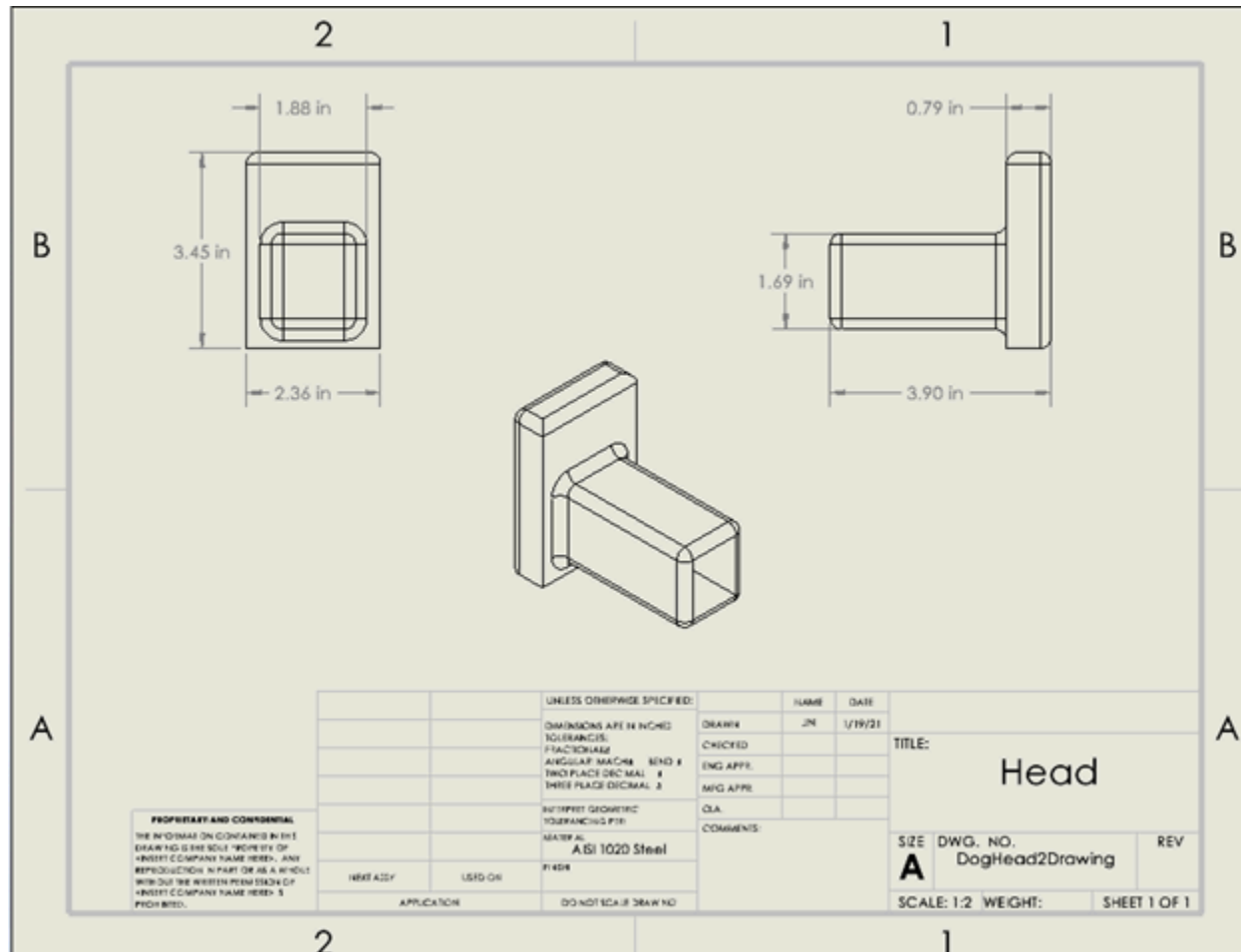


Figure 24. Dog Prototype – Head Drawing – 1 Total

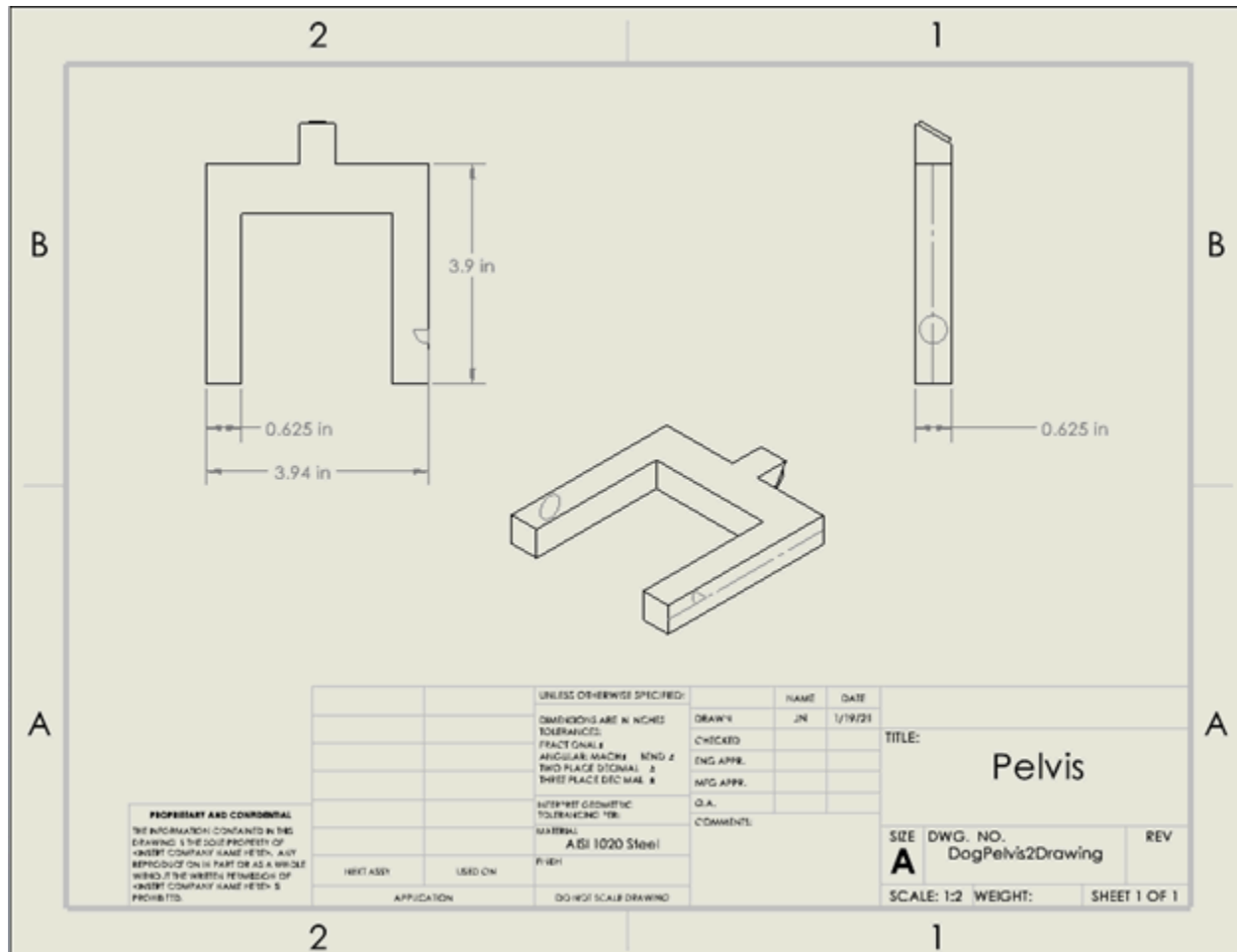


Figure 25. Dog Prototype – Pelvis Drawing – 1 Total

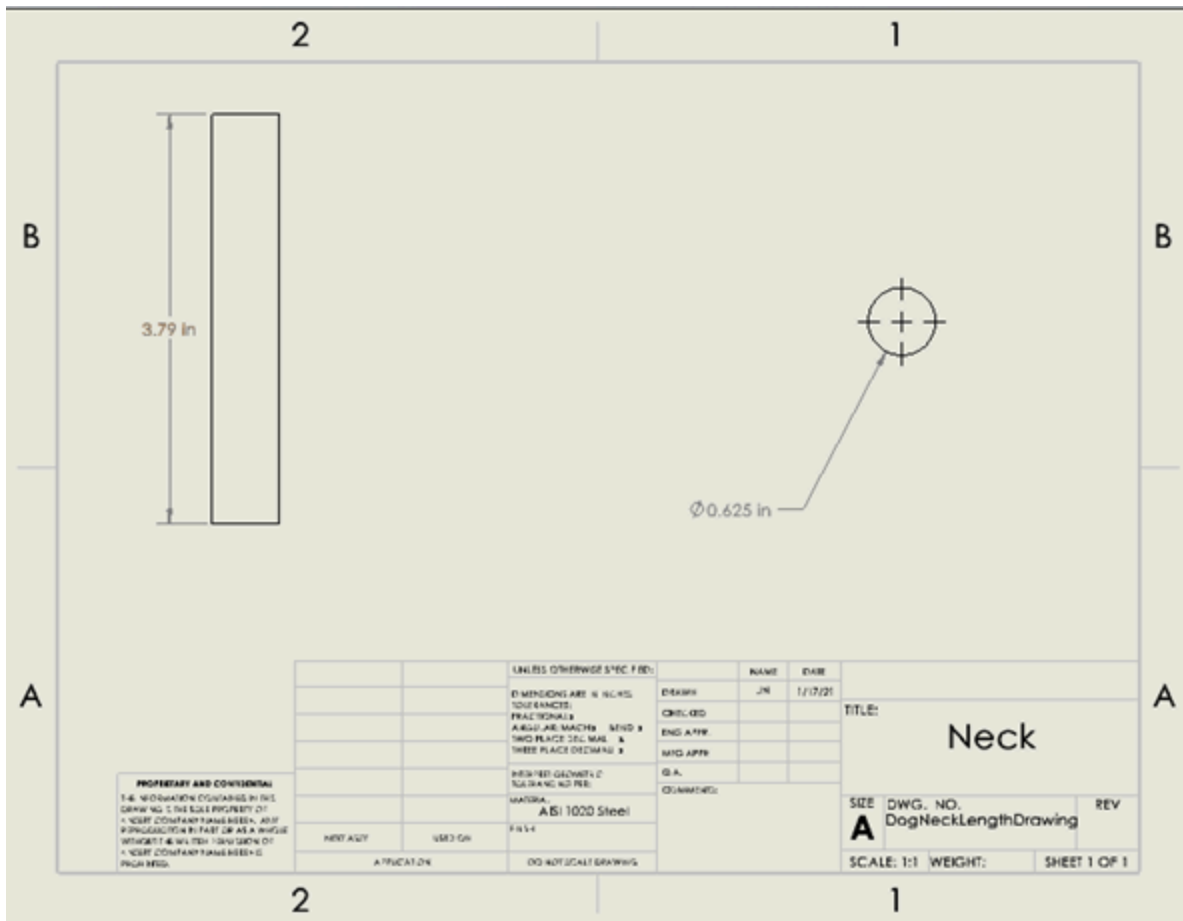


Figure 26. Dog Prototype – Neck Drawing – 1 Total

Complete Design

The drawing of the complete design features a table of all components as well as the necessary quantity for each. The parts are also labeled on the dog and are easily identifiable by number. The material is also stated, and the document is initialed and dated by the Design Lead Jake Nickoloff.

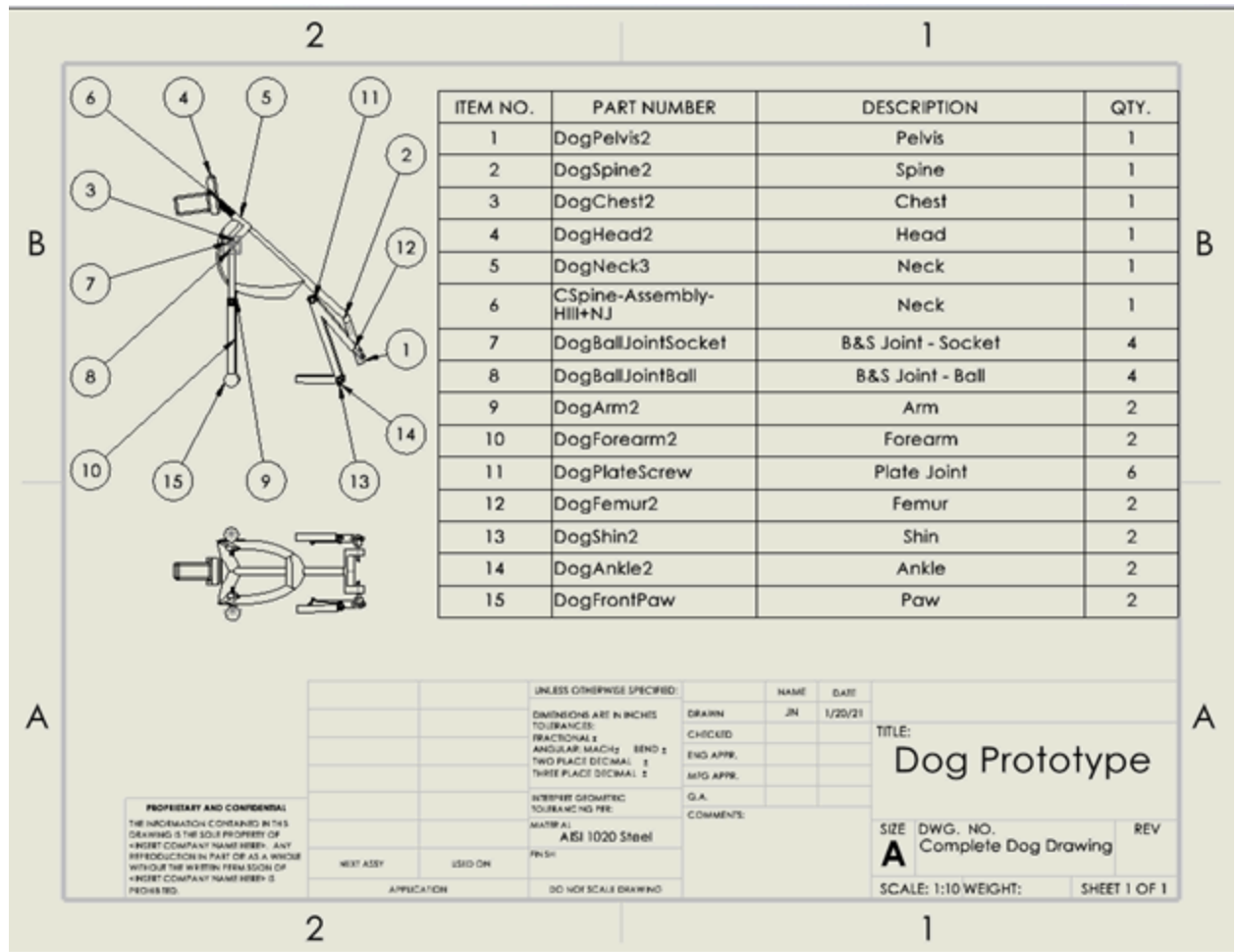


Figure 27. Dog Prototype – Complete Dog Drawing

Testing and Validation

The team is expecting a minimum of 3 different tests to be performed on the dummy. Since there are three materials that will be used to build the canine dummy, the team will focus on doing the tests that experiment the dummy's as a whole. Three materials (metal, foam and fabric) combined will act as one composite material which has its own properties.

An important requirement from the customers is the dummy will have a center of mass that is similar to that of an actual medium size Beagle. Thus, the first test will focus on determining the center of mass of the dummy. The team will hang the dummy up from a random

point and use a string to draw a line on the dummy straight down from the hanging point. Then, the dummy will be hung again but from another random location and the team will repeat the line drawing steps. The intersection between the two lines will be the dummy's center of mass. Similarly, team members will repeat this process for all three directions and thus, the final center of mass is a vector consisting of the measured x, y, and z values.

Another required function is the dummy will not break when the car is moving at 30 mph and then, suddenly stopped. To mimic such an experiment, the team will perform a drop test with a purchased harness. The dummy will be dropped from different heights, and team members will observe and record the behavior of the dummy's legs and neck. In addition, the dummy should be able to withstand a high drop without breaking. Force calculation will be needed to determine the highest point the dummy shall be dropped to reach 30 mph before stopping. Also, a high-speed camera will be used to capture the moment when the harness stops the dummy and compression causes deflection onto the dummy's chest. With this information, the team will be able to observe and analyze how all three materials (steel, foam and fabric) will work together as a whole. Based on the result, the overall dummy's composite material's strengths can be determined.

The last experiment is a configuration test where the team will configure the dummy into two different positions, sitting and laying down. Then, team members will measure the width, height, and length of the dummy to verify the prototype's dimensions with the designed dimensions. In addition, the team will also clarify whether the dummy will be in the sitting or laying down position by itself, meaning it will stay still without further support from any other means.

All of the above experiments will be performed using the tools and equipment available in the engineering shop and Capstone lab. The center of mass test will be repeated for a minimum 3 times as the dummy will be dropped at different heights. In addition, multiple drops are needed to determine the best angle and location to fix the high-speed camera. Overall, all tests are expected to be repeated multiple times as modifications occur throughout the fabrication and testing processes.

Timeline

The project is 60% completed with both Fall 2020 and Spring 2021 taken into consideration. The team has been working on the project for 18 weeks out of the allowed 30 weeks. Overall, the project is on the schedule; however, the team should input more time and effort into fabricating the dummy at the beginning of the Spring semester as to leave enough time for final reports, presentations, and showcase preparations in April 2021. According to the current plan (see Appendix A), team members have entered the fabrication phase of the dummy's bone structure and are expected to complete this phase by the end of week 20 (out of 30 weeks). In addition, the team is scheduled to purchase canine car harnesses, foam, fabric and metal balls for the dummy's feet in week 19. Week 21 is unplanned as to prepare for any unpredicted delays in schedule.

From week 22 to 24, the team will also begin preparing and adding the foam on the bone structure and sewing canvas fabric to make a "suit" (skin) for the dummy. In week 25, team members will enter the testing stage where the team will perform center of mass, configuration, and drop tests with the final assembled dummy including bone structure (bone), foam (muscle), and canvas fabric (skin). This stage is scheduled to last for 3 weeks as the team will simultaneously fabricate the dummy, perform tests and make modifications. The team will focus

on final reports, presentations, and engineering showcases during the last week. Responsibility of each member and percent complete of each task are color-coded and represented in the timeline.

Budget

Table # displays information of the canine crash test dummy expenses so far. As of today, the team has spent \$409.09, including taxes. Therefore, the team still has \$3,590.91 remaining to spend on the rest of the materials on the bill of materials and services for certain parts of the canine crash test dummy.

Table #: Expenses Information

Date of Purchase	Order #	Quantity	Material	Cost	Supplier
11/18/2020	167816	1	Sample Request	\$5.95	Big Duck Canvas
12/2/2020	6766857	1	12-piece foam material sample pack	\$47.47	Mc Master
12/2/2020	6766857	1	6-piece foam hardness sample pack	\$32.97	Mc Master
12/2/2020	6766857	1	Shipping and tax	\$15.41	Mc Master
01/13/2021	4920615	1	CR Steel Sheet 16 GA (36" x 60")	\$73.63	Industrial Metal Supply Co.
01/13/2021	4920615	1	1018 CR Steel Rnd 5/8 x 10FT ASTM A 108 (60")	\$18.37	Industrial Metal Supply Co.
01/13/2021	4920615	1	Steel Tube Sq 1 1/2 X 1 1/2 x .250 ASTM A 500 (60")	\$62.20	Industrial Metal Supply Co.
01/13/2021	4920615	1	1018 CR Steel Flat Bar R/L 5/8 X 3 ASTM A 108 (60")	\$96.88	Industrial Metal Supply Co.
01/13/2021	4920615	1	Shipping and tax	\$21.59	Industrial Metal Supply Co.
01/15/2021	485	1	Flat Washer Zinc 3/8 25PK	\$3.98	The Home Depot
01/15/2021	485	1	Hex Nut Zinc 3/8 x 25PK	\$3.56	The Home Depot
01/15/2021	485	1	Hex Bolt Zinc 3/8 X1	\$2.40	The Home

					Depot
01/15/2021	485	1	1/2 Drive Universal Joint	\$21.94	The Home Depot
01/15/2021	485	1	Shipping and tax	\$2.74	The Home Depot

The team reached out to Metal Supermarkets Phoenix (southwest) to have them bend the 16 gauge metal sheet for the chest cavity of the canine crash test dummy. The team decided to outsource the chest cavity of the dummy since Grand Canyon University's engineering department does not have the appropriate tools to bend 16 gauge metal sheets to the degree the team desires. Metal Supermarkets Phoenix (southwest) service will fall on the remaining budget. The team however has not been able to get a quote from them. Once a quote is provided, the budget will be updated. Table # provides the materials that the team is planning on purchasing at the beginning of next week, January 25th.

Table #: Remaining Purchases

Quantity	Material	Cost	Supplier
1	24" x 24" x 1/2" Polyurethane Foam Sheet	\$20.00	Mc Master
1	8 oz. Gorilla Glue	\$11.00	The Home Depot
1	60" x 36" 10 oz. Canvas Cloth	\$8.16	Big Duck Canvas
1	Shipping and Taxes	\$25.00	

The remaining materials total cost is \$64.16, bringing up the budget expenses to \$473.25, including taxes. The team plans to have a \$550 contingency budget in case anything goes completely wrong, meaning that the team will be able to purchase any parts that get damaged

through the assembling process. Therefore, the total expenses would be \$1,023.25, leaving a total of \$2,976.75 remaining. If time allows, the team would want to construct another size of a canine crash test dummy using the remaining budget.

Conclusion - Christian

The goal is to have a functioning prototype fully manufactured by the end of the semester. The prototype must meet all functional requirements so that it can be used in the ASTM committee's second proposal to have standards established for canine harness crash testing. The team will continue to utilize all of the standards mentioned above as guidelines throughout the rest of the project's progression. As of right now the team will continue to fabricate the metal bone structure of the dummy to the specifications of the solidworks drawing included with intentions of having it finished by week 20 as according to the project timeline. After that, the remaining materials listed in the budget section will be purchased and the fabrication of the foam and canvas layers will commence. If the manufacturing process remains on schedule as it is expected to, the prototype will be complete by week 24 at the latest which will then mark the beginning of the testing verification phase of the project. The tests mentioned previously will be performed on the fully assembled prototype. Once the product passes all the tests according to the testing standards that are established, it will be ready to be presented in the capstone showcase.

References

American Nurses Association. (n.d.). *Scope of practice*.

<https://www.nursingworld.org/practice-policy/scope-of-practice/>

American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.). <https://doi.org/10.1037/0000165-000>

ASTM D2651-01(2016), Standard Guide for Preparation of Metal Surfaces for Adhesive Bonding, ASTM International, West Conshohocken, PA, 2016, www.astm.org

ASTM D3574-17, Standard Test Methods for Flexible Cellular Materials—Slab, Bonded, and Molded Urethane Foams, ASTM International, West Conshohocken, PA, 2017, www.astm.org

ASTM D3884-01, Standard Guide for Abrasion Resistance of Textile Fabrics (Rotary Platform, Double-Head Method), ASTM International, West Conshohocken, PA, 2001, www.astm.org

Center for Substance Abuse Treatment. (2014). *Improving cultural competence* (HHS Publication No. 14-4849). U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration. <https://www.ncbi.nlm.nih.gov/books/NBK248428/>

Copeland, T., Henderson, B., Mayer, B., & Nicholson, S. (2013). Three different paths for tabletop gaming in school libraries. *Library Trends*, 61(4), 825–835. <https://doi.org/10.1353/lib.2013.0018>

Holland, R. A., & Forrest, B. K. (2017). *Good arguments: Making your case in writing and public speaking*. Baker Academic.

U.S. Department of Transportation. (2014). *Laboratory Test procedure for FMVSS 213 Child Restraint systems*. Retrieved from

https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/documents/tp-213-10_tag.pdf

U.S. Department of Transportation. (1994). *Laboratory Test Procedure for FMVSS 210 Seat Belt Assembly Anchorages*. Retrieved from

[https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/documents/tp-210-09a_tag.pdf#:~:text=FMVSS%20210%20establishes%20requirements%20for,MPVs\)%2C%20trucks%20and%20buses.](https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/documents/tp-210-09a_tag.pdf#:~:text=FMVSS%20210%20establishes%20requirements%20for,MPVs)%2C%20trucks%20and%20buses.)

U.S. Department of Transportation. (2007). *Laboratory Test Procedure for FMVSS 209 Seat Belt Assemblies*. Retrieved from

https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/documents/tp-209-08_tag.pdf#:~:text=FMVSS%20209%2C%20Seat%20Belt%20Assemblies,person%20at%20any%20one%20time.

Appendix

Bill of Materials

Materials	Quantity	Cost	Supplier
Sample Request	1	\$5.95	Big Duck Canvas
12-piece foam material sample pack	1	\$47.47	Mc Master
6-piece foam hardness sample pack	1	\$32.97	Mc Master
Shipping and tax	1	\$15.41	Mc Master
CR Steel Sheet 16 GA (36" x 60")	1	\$73.63	Industrial Metal Supply Co.
1018 CR Steel Rnd 5/8 x10FT ASTM A 108 (60")	1	\$18.37	Industrial Metal Supply Co.
Steel Tube Sq 1 1/2 X 1 1/2 x .250 ASTM A 500 (60")	1	\$62.20	Industrial Metal Supply Co.
1018 CR Steel Flt Bar R/L 5/8 X 3 ASTM A 108 (60")	1	\$96.88	Industrial Metal Supply Co.
Shipping and tax	1	\$21.59	Industrial Metal Supply Co.
Flat Washer Zinc 3/8 25PK	1	\$3.98	The Home Depot
Hex Nut Zinc 3/8 x 25PK	1	\$3.56	The Home Depot
Hex Bolt Zinc 3/8 X1	1	\$2.40	The Home Depot
1/2 Drive Universal Joint	1	\$21.94	The Home Depot
Shipping and tax	1	\$2.74	The Home Depot
24" x 24" x 1/2" Polyurethane Foam Sheet	1	\$20.00	Mc Master
8 oz. Gorilla Glue	1	\$11.00	The Home Depot
60" x 36" 10 oz. Canvas Cloth	1	\$8.16	Big Duck Canvas

Shipping and Taxes	1	\$25.00	
Contingency Plan	\$550		
Total Cost	\$1,023.25		